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Comparative evaluation of deep learning models for lung segmentation in chest X-rays: applications in infectious disease screening

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Abstract

Respiratory infections such as pneumonia and tuberculosis remain the leading contributors to global morbidity and mortality, and early diagnosis is essential for effective treatment. Chest X-ray imaging is widely used for the detection of these conditions; however, reliable interpretation requires accurate delineation of lung fields. Manual annotation is time-consuming and prone to variability, underscoring the need for automated segmentation methods. In this study, we conducted a systematic evaluation of state-of-the-art deep learning architectures for binary lung segmentation, including U-Net, Attention U-Net, Double U-Net, U2-Net, VGG-UNet, UNet++, ResNet-UNet, Dense-UNet, Swin U-Net and HieraSeg Net. The performance of the model was compared using the dice coefficient, the intersection of the union (IoU), the mean absolute error (MAE), the Hausdorff distance and the average symmetric surface distance (ASSD). Among these models, Dense-UNet achieved the best results, yielding a Dice score of 0.9848 and an IoU of 0.9700, with the lowest surface error measures. These findings highlight the potential of Dense-UNet to serve as a robust backbone for AI-assisted diagnostic systems in infectious respiratory diseases, thereby supporting faster, more reliable, and scalable approaches to clinical decision-making.

Keywords Infectious disease screening, Lung segmentation, Chest radiographs, U-Net, Convolutional neural networks, Medical image analysis

Introduction

Lung segmentation is a fundamental task in medical image analysis, particularly in supporting radiologists for the detection and diagnosis of pulmonary and infectious respiratory diseases such as pneumonia, tuberculosis, and lung cancer. With the widespread adoption of imaging modalities such as chest X-rays (CXRs) and computed tomography (CT) scans in clinical practice, the demand for efficient and accurate lung segmentation tools has increased substantially. Accurate delineation of lung fields is especially critical in the context of infectious disease screening, where rapid and reliable interpretation can directly impact treatment decisions and public health outcomes. However, lung segmentation

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remains a challenging task due to the inherent variability in the shape, size, and texture of lung regions, as well as the presence of complex structures such as nodules, lesions, and infection-induced opacities. Recent advancements in deep learning, particularly convolutional neural networks (CNNs), have led to substantial progress in medical image segmentation. Among these, U-Net has emerged as one of the most widely adopted architectures in biomedical imaging, owing to its symmetric encoder–decoder design. Despite its success, U-Net and its derivatives often encounter difficulties in delineating boundaries, especially when segmenting small or irregularly shaped regions associated with infectious diseases. To address these limitations, researchers have introduced enhanced models incorporating attention mechanisms and multi-scale feature learning.

Recent advances in deep learning have led to the development of a wide range of convolutional neural network–based architectures for lung segmentation in chest radiographs. These methods have substantially improved segmentation accuracy and enabled more reliable analysis of pulmonary structures in infectious disease screening. However, challenges related to boundary precision, robustness to infection-induced variations, and computational efficiency continue to limit their broader clinical adoption. To address these issues, a systematic evaluation of representative segmentation architectures is necessary to better understand their practical strengths and limitations.

To address these challenges, a wide range of architectural enhancements have been proposed, including attention mechanisms, dense and residual connections, multi-scale feature learning, and hybrid designs that combine convolutional and transformer-based components. Although these approaches demonstrate promising results, their relative strengths and limitations are often evaluated in different experimental settings, making it difficult to draw consistent conclusions regarding their suitability for the screening of infectious diseases in the real-world.

Motivated by these observations, this study presents a comprehensive and systematic evaluation of state-of-the-art deep learning architectures for binary lung segmentation in chest X-ray images. The objective is not to introduce a new segmentation model, but to benchmark representative architectures under a unified experimental framework, with particular emphasis on robustness, boundary accuracy, and clinical relevance. By comparing conventional U-Net variants with advanced architectures, this work aims to identify segmentation backbones that offer an effective balance between accuracy, efficiency, and reliability, thereby supporting the development of scalable AI-driven diagnostic pipelines for infectious respiratory diseases.

The aim of this study is to conduct a systematic evaluation of multiple deep learning-based segmentation models for lung field extraction from chest X-ray images, with a focus on applications in infectious respiratory disease detection. Specifically, we compare conventional U-Net [1], Attention U-Net, and advanced variants such as Dense-UNet and U^2-Net, using metrics that assess both segmentation accuracy and boundary precision. Performance is measured through Dice Coefficient, Intersection over Union (IoU), Hausdorff Distance, and Average Symmetric Surface Distance (ASSD). By benchmarking these models, this work seeks to identify architectures that balance accuracy, efficiency, and robustness, thereby contributing to the development of clinically applicable tools for automated lung segmentation in infectious disease screening. The major contributions of this study are as follows:

- A **comprehensive evaluation of ten state-of-the-art deep learning architectures** for binary lung segmentation in chest X-ray images is presented, covering both conventional U-Net variants and recent transformer-based approaches.
- Model performance is benchmarked using **multiple complementary metrics**—Dice Coefficient, Intersection over Union (IoU), Mean Absolute Error (MAE), Hausdorff Distance, and Average Symmetric Surface Distance (ASSD)—to enable a rigorous and holistic comparison.
- Experimental findings demonstrate that **Dense-UNet achieved superior segmentation accuracy** and lower surface errors compared to other models, confirming its effectiveness for lung boundary delineation in chest X-rays.
- The study underscores the **clinical relevance of accurate lung segmentation** as a foundation for AI-driven diagnostic systems, particularly in the detection, monitoring, and management of **infectious respiratory diseases such as pneumonia, tuberculosis, and COVID-19**.
- A **scalable framework** is provided that can be extended to multi-class segmentation and adapted for real-world deployment in infectious disease screening and broader healthcare applications.

The remainder of this paper is organized as follows. Section "Related Work" reviews prior work on lung segmentation using convolutional and hybrid deep learning architectures. Section "Proposed Methodology" details the proposed methodology, including the collection, preprocessing, augmentation, and implementation of different segmentation models. Section "Experimental Results" presents the experimental evaluation, highlighting both quantitative and qualitative results. Section "Conclusion

and Future Scope" provides a detailed discussion of the findings, emphasizing architectural trade-offs and clinical relevance. Finally, the last section concludes the paper and outlines directions for future research.

Related work

This section reviews key advancements in medical image segmentation, with a focus on lung segmentation using CNN architectures and hybrid deep learning models. For example, Wahyuningrum et al. [2] developed DANet, which employs a hybrid attention mechanism to refine segmentation boundaries, achieving a Dice score of 0.9729. However, their study also highlighted the challenges associated with hyperparameter optimization and computational overhead. Similarly, Zhong et al. [3] proposed EF-Net for COVID-19 CT scan segmentation, obtaining a Dice score of 98.1% but reporting significant performance degradation on noisy data. Several other modifications to U-Net have sought to strengthen multi-scale feature extraction. Furthermore, although hybrid models such as Guide-YNet [4] demonstrate high accuracy, their increased architectural complexity raises concerns regarding computational efficiency and scalability. Another key limitation is the lack of clinical validation, as relatively few studies assess model performance in real-world infectious disease screening or incorporate physician-verified outcomes.

Attention mechanisms have been widely adopted to improve boundary delineation in segmentation tasks. To further address boundary blurring, Liu et al. [5] developed CSEA-Net, which used spatial channel attention to reach a Dice score of 98.65% in the LNDb data set, although the authors noted the need for clinical validation. Collectively, these works demonstrate the potential of attention mechanisms for boundary refinement, but also highlight the need for robustness and validation in real-world clinical environments.

U-Net variants and multi-scale feature learning

Numerous U-Net variants have been proposed to enhance multi-scale feature extraction. Lu et al. [6] developed ParaU-Net, which integrates a Multi-Scale Parallel Encoding Module (MPEM) and achieved a Dice score of 92.16%, although challenges remained to balance local and global contextual information. Lin et al. [7] proposed MUS-Net to address uncertainty in nodule segmentation, reporting a Dice score of 91.20%, but their study did not explore clinical translation. Arvind et al. [8] incorporated dropout layers into U-Net for CXR segmentation, mitigating overfitting and achieving a Dice score of 91.87%. Similarly, Ilhan et al. [9] applied histogram-based preprocessing for COVID-19 CT segmentation, achieving 85% Dice. These results underscore that while U-Net variants improve feature learning, challenges remain in ensuring generalizability and clinical applicability.

Hybrid architectures and transformers

Hybrid architectures combining CNNs, multi-modal data, and transformer-based modules have gained attention for their superior representational power. Zhou et al. [4] introduced Guide-YNet, a dual-encoder U-Net that fuses PET/CT data, achieving a Dice score of 97.77%. Chaoyang et al. [10] proposed FDR-TransUNet, integrating Vision Transformers with Feature Double Reuse blocks, which reached 97.5% Dice. Despite their high accuracy, computational demands were not quantified, raising concerns about scalability. Other studies have explored different hybrid strategies, such as Vincy et al. [11], who combined 3D ResNet50 with U-Net, and Owais et al. [12], who introduced DMDF-Net, achieving a Dice score of 75.7%. These studies highlight both the promise and limitations of hybrid approaches, especially in the treatment of small lesions.

Segmentation based approaches in the respiratory diseases

Recent research shows that segmentation-driven pipelines play a vital role in respiratory disease detection by isolating lung and infection regions before diagnosis. Several studies have demonstrated that incorporating lung segmentation into chest X-ray analysis improves robustness when detecting COVID-19, pneumonia, and tuberculosis, particularly in heterogeneous clinical datasets. Infection-level segmentation has further enabled quantitative assessment of disease spread, supporting severity estimation rather than coarse image-level predictions [13, 14].

Key insights and gaps

While attention-based models such as EF-Net [3] and CSEA-Net [5] perform well on clean datasets, they often falter on noisy or heterogeneous inputs. Clinical validation remains limited in works such as DANet [2] and MUS-Net [7], restricting real-world applicability. Furthermore, although hybrid models like Guide-YNet [4] and FDR-TransUNet [10] deliver high segmentation accuracy, their computational complexity poses challenges for practical deployment. These gaps underscore the need for architectures that achieve a balance between accuracy, robustness, and efficiency for lung segmentation in clinical practice.

Proposed methodology

Segmentation of lung regions from chest radiographs involves a sequence of pre-processing operations followed by the application of deep learning models to

assess segmentation performance. This section outlines the methodology adopted in this study.

Dataset collection

A clearly defined methodology is essential in medical image analysis to ensure reproducibility and facilitate objective evaluation. For this study, the publicly available *Chest X-ray Masks and Labels* dataset was employed [15]. The dataset comprises 704 grayscale chest X-ray images with varying resolutions and patient positioning. Each image is paired with a corresponding binary mask that delineates the lung regions. These masks were annotated by expert radiologists and serve as the ground truth for model training and evaluation. To ensure a structured training process, the dataset was partitioned into training and validation subsets, comprising 563 and 141 images, respectively, following an 80:20 split. Representative examples of chest X-ray images and their corresponding ground-truth masks are shown in Fig. 1.

Data preprocessing

To maintain uniformity across all input samples and optimize them for neural network processing, a series of preprocessing techniques were employed. Initially, all images and their corresponding binary masks were resized to dimensions of $256 \times 256 \times 1$ using bilinear interpolation. Formally, the resized pixel value $I\prime(x, y)$ at a coordinate (x, y) is computed as a weighted average of the surrounding pixels:

$$I'(x, y) = \sum_{i=0}^{1} \sum_{j=0}^{1} \omega_{ij} \, I(x_i, y_j), \tag{1}$$

where ω_{ij} are the interpolation weights based on the distance from (x, y).

Subsequently, normalization was applied to scale pixel intensity values into the range $[0, 1]$ using Min-Max normalization [16]. Given an original pixel value $I(x, y)$, the normalized value $I_{\text{norm}}(x, y)$ is calculated as:

$$I_{\text{norm}}(x, y) = \frac{I(x, y) - I_{\min}}{I_{\max} - I_{\min}}, \tag{2}$$

where $I_{\min}$ and $I_{\max}$ represent the minimum and maximum intensity values in the image, respectively. This standardization ensures consistency across the dataset, facilitates stable model training by preventing large

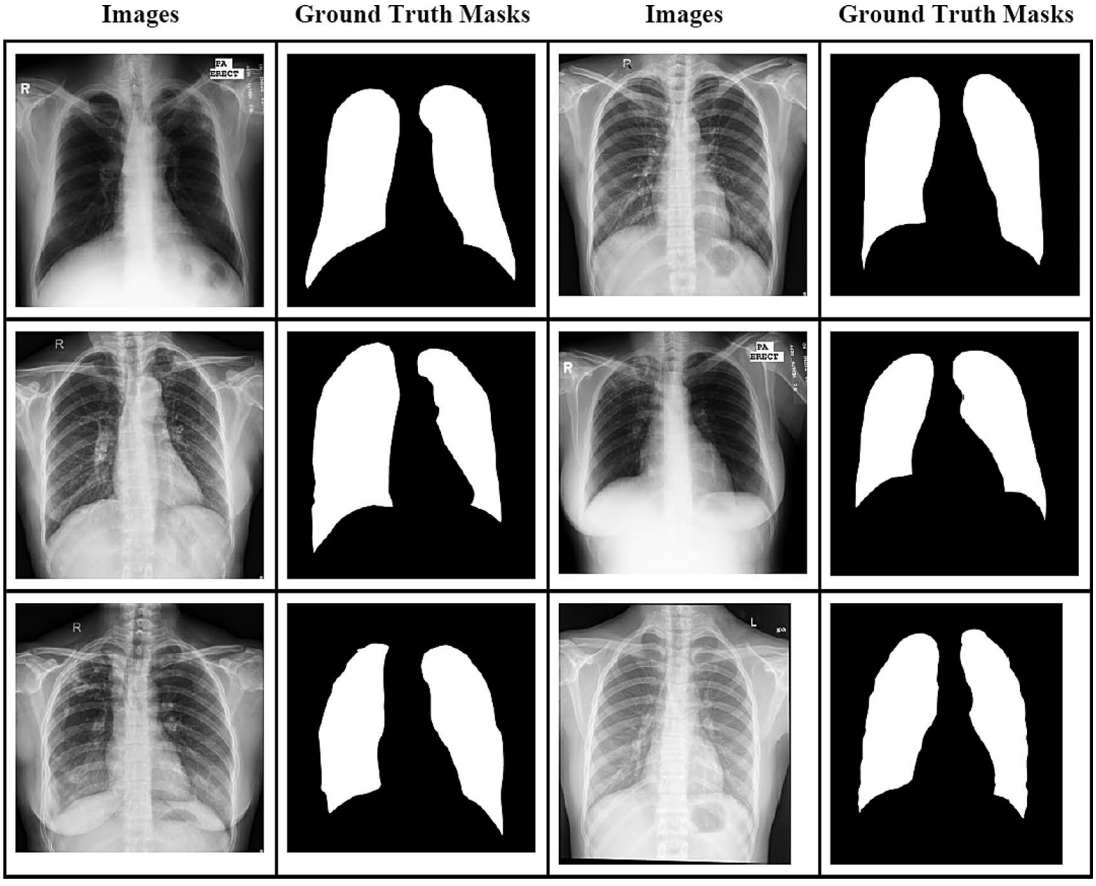


Fig. 1 Representative chest X-ray images along with their corresponding ground-truth masks from the dataset

gradients, and improves convergence speed while mitigating numerical instability.

In addition to resizing and normalization, the preprocessing pipeline incorporated further enhancements. Contrast-limited adaptive histogram equalization (CLAHE) [17] was applied to the images using the `cv2.createCLAHE()` function with a clip limit of 2.0 and a tile grid size of 8×8. CLAHE operates by dividing the image into small regions and applying histogram equalization with a contrast limiting threshold T_{clip}, thus preventing noise amplification in homogeneous regions. Furthermore, Non-Local Means (NLM) denoising was applied to reduce noise and artifacts, utilizing the `cv2.fastNlMeansDenoising()` function. The NLM algorithm estimates the denoised pixel value $I_d(x)$ as a weighted average of similar pixels across the image:

$$I_d(x) = \frac{1}{C(x)} \sum_{y \in \Omega} \omega(x, y)\, I(y), \tag{3}$$

where $\omega(x, y)$ measures the similarity between patches around x and y, $C(x)$ is a normalizing factor and Ω denotes the search window. This method effectively smooths out noise while preserving important edge information, which is critical for accurate segmentation.

Data augmentation

Data augmentation techniques were implemented using the `Albumentations` library to artificially enhance the diversity of the training dataset, thereby improving model robustness and generalization. The training data set was augmented with an augmentation factor of 1, ensuring that each image in the training set had an augmented counterpart. These augmentations simulate variations in patient positioning, image quality, and potential pathologies, thus improving the model's ability to generalize to unseen data. The augmentation pipeline included horizontal and vertical flips, which introduced spatial variations that simulate different patient orientations. Random brightness and contrast adjustments were applied to mimic variations in image acquisition conditions. Furthermore, elastic transformations and grid distortion were used to simulate subtle anatomical deformations and local warping effects, respectively, while Gaussian noise addition improved the model's resilience against noisy or low-quality images.

Model architectures

This study investigates the application of ten different CNN-based architectures for automated segmentation of lung regions in chest radiograph images. All models were implemented using `TensorFlow` and `Keras`, building upon the foundational U-Net design with architectural modifications aimed at enhancing segmentation performance. These architectures were chosen to explore the impact of different architectural features, such as attention, residual connections, dense connections, and transformer-based encoding, on the accuracy and robustness of lung segmentation.

U-Net

The base U-Net architecture consists of an encoder-decoder network with skip connections that preserve spatial information between corresponding layers [18]. Let the input image be denoted as $X \in \mathbb{R}^{H \times W \times C}$. The encoder progressively applies convolution and downsampling operations to extract hierarchical features, which can be expressed as:

$$F^l = f^l_{\text{enc}}(F^{l-1}), \quad l = 1, 2, \ldots, L \tag{4}$$

where $F^0 = X$, f^l_{enc} represents the convolutional block at layer l in the encoder, and L is the number of encoding layers. Each encoder layer captures increasingly abstract representations of the input while reducing spatial resolution. The decoder reconstructs the feature maps back to the original resolution using upsampling operations:

$$\hat{F}^l = f^l_{\text{dec}}(\text{Upsample}(\hat{F}^{l+1}) \oplus F^l), \quad l = L-1, \ldots, 0 \tag{5}$$

where $\hat{F}^l$ is the feature map at decoder layer l, $\text{Upsample}(\cdot)$ denotes the upsampling operation (e.g., transposed convolution), $\oplus$ represents the concatenation with the corresponding encoder feature map (skip connection), and f^l_{dec} is the convolutional block in the decoder. The final output $\hat{Y} = \hat{F}^0$ represents the predicted segmentation mask. A diagrammatic representation of the U-Net architecture is shown in Fig. 2, illustrating the encoder-decoder structure and skip connections.

Attention U-Net

The Attention U-Net model extends the standard U-Net architecture by incorporating attention mechanisms, which allow the network to focus on salient regions of the feature maps while suppressing irrelevant or noisy areas [19]. This is particularly useful for improving the accuracy of lung boundary delineation in chest X-ray images. Let the feature map from the encoder at layer l be F^l and the gating signal from the decoder at layer $l+1$ be G^{l+1}. The attention gate computes an attention coefficient map α^l as:

$$\alpha^l = \sigma\big(W_\psi^T \,\text{ReLU}(W_F F^l + W_G G^{l+1} + b)\big), \tag{6}$$

where W_F, W_G, and W_ψ are learnable weight matrices, b is a bias term, σ denotes the sigmoid activation, and $\text{ReLU}(\cdot)$ is the rectified linear unit. The attention map α^l

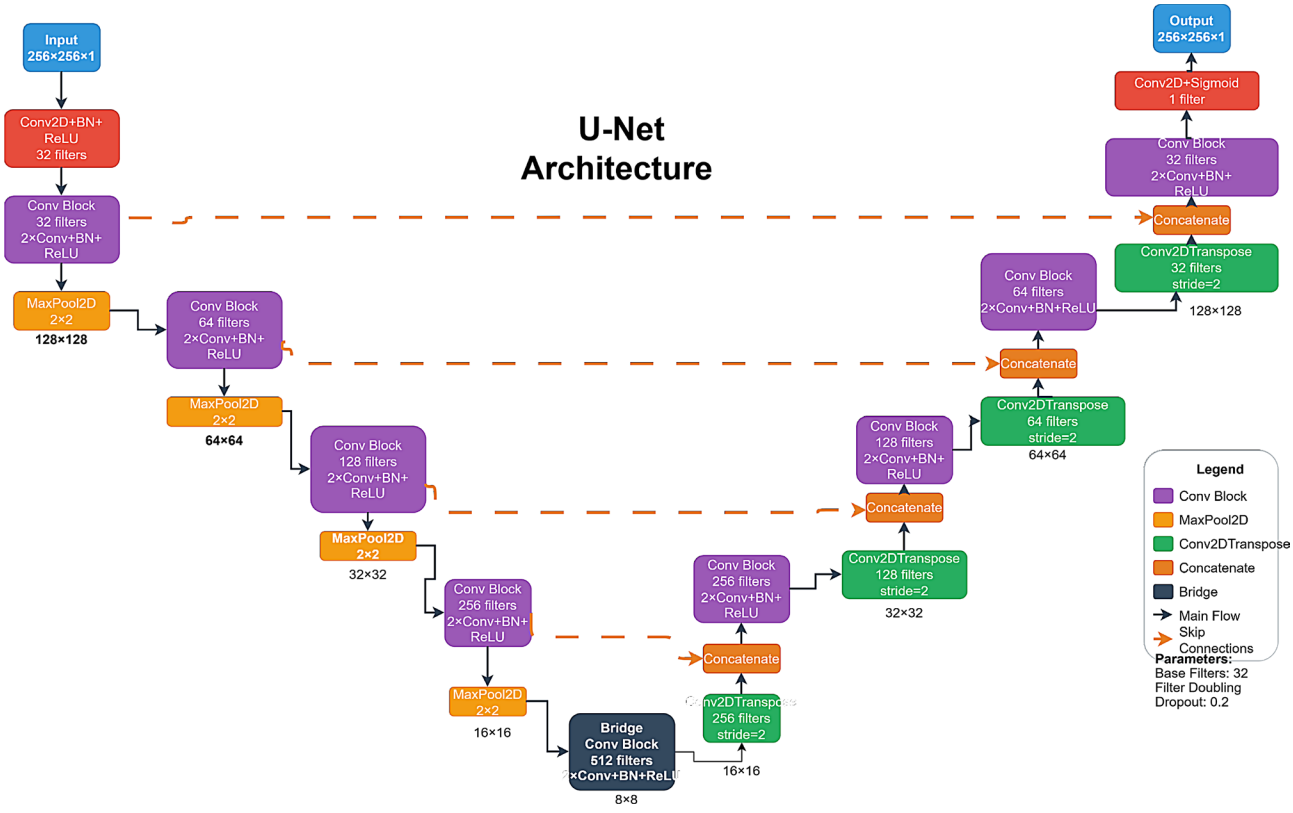


Fig. 2 Diagrammatic representation of U-Net architecture, showing encoder-decoder layers and skip connections

is then applied to the encoder feature map to obtain the refined features:

$$\tilde{F}^l = \alpha^l \odot F^l, \tag{7}$$

where $\odot$ represents element-wise multiplication. The refined feature map $\tilde{F}^l$ is concatenated with the upsampled decoder features as in the standard U-Net, and the decoder proceeds with reconstruction of the segmentation mask. Figure 3 illustrates the architecture of Attention U-Net, showing how attention gates are integrated into the skip connections to selectively emphasize relevant features during decoding.

U²-Net

U^2-Net is another architecture employed in this study, characterized by its nested U-structure design. Unlike the standard U-Net, which follows a single encoder-decoder path, U^2-Net incorporates multiple Residual U-blocks (RSUs) at different scales [20]. Each RSU itself follows a mini U-Net structure with residual connections, allowing the model to capture both local fine-grained features and broader global contextual information. This hierarchical design enhances the network's ability to segment objects with varying sizes and irregular shapes, which is particularly advantageous for delineating lung regions in chest X-ray images. Formally, let the input feature map be denoted as $X \in \mathbb{R}^{H\times W\times C}$. An RSU block begins with an initial convolutional transformation:

$$F_0 = f_{\text{in}}(X), \tag{8}$$

where $f_{\text{in}}(\cdot)$ is a convolutional operator. The encoder part of the RSU performs progressive downsampling to extract multi-scale features:

$$F^l = f^l_{\text{enc}}(F^{l-1}), \quad l = 1, 2, \ldots, L, \tag{9}$$

while the decoder reconstructs the feature maps via upsampling and skip connections:

$$\hat{F}^{l-1} = f^l_{\text{dec}}(\text{Upsample}(\hat{F}^l) \oplus F^{l-1}), \quad l = L, L-1, \ldots, 1. \tag{10}$$

Finally, the output of the RSU block is refined using a residual connection:

$$Y = F_0 + \hat{F}^0, \tag{11}$$

where $\oplus$ denotes concatenation and Y is the output feature map. The residual formulation improves gradient flow during training and encourages feature reuse. By stacking multiple RSU blocks within a larger U-shaped

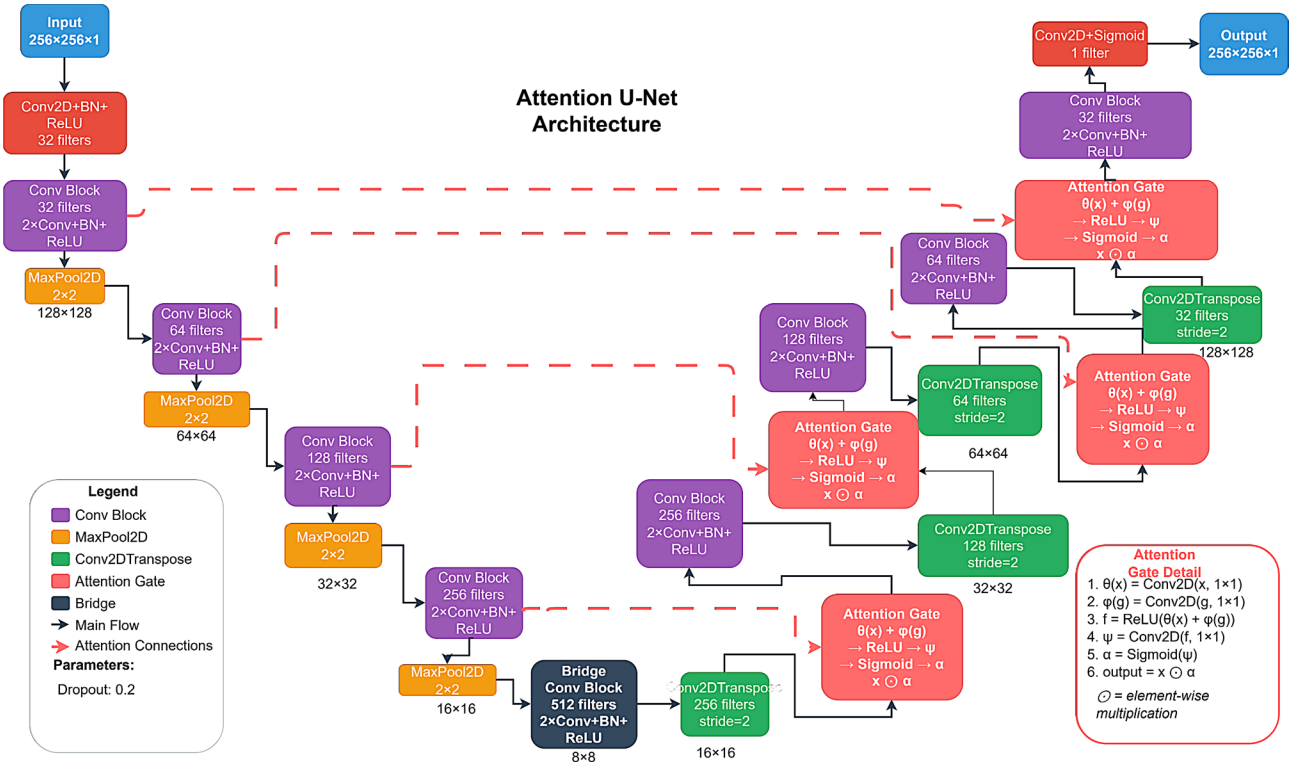


Fig. 3 Schematic representation of attention U-Net architecture. Attention gates are integrated into the skip connections to selectively focus on relevant regions

framework, U^2-Net captures multi-scale information more effectively than conventional U-Net variants.

Double U-Net and U-Net++

Double U-Net and U-Net++ are advanced variants of the original U-Net [21], designed to enhance feature representation and improve segmentation accuracy in challenging medical imaging tasks.

Double U-Net. The Double U-Net architecture consists of two U-Net models stacked sequentially. The first U-Net performs an initial coarse segmentation, and its output feature maps are passed to the second U-Net for further refinement [22]. This cascaded design enables the model to correct errors from the first stage and progressively enhance boundary delineation.

Formally, let X denote the input image and $U_1(\cdot), U_2(\cdot)$ represent the encoder-decoder mappings of the first and second U-Nets, respectively. The intermediate prediction from the first U-Net is given by:

$$S_1 = U_1(X), \tag{12}$$

which is concatenated with the original input and fed into the second U-Net:

$$S_2 = U_2(X \oplus S_1), \tag{13}$$

where $\oplus$ denotes concatenation. The final segmentation mask is obtained from S_2, which typically exhibits improved structural consistency compared to S_1.

U-Net++. U-Net++ [23] introduces nested and dense skip pathways between the encoder and decoder subnetworks. In contrast to the simple skip connections in standard U-Net, U-Net++ employs a series of intermediate convolutional layers at each skip level, which progressively refine the transmitted feature maps. This design reduces the semantic gap between encoder features (low-level) and decoder features (high-level), leading to more precise segmentation.

Mathematically, let $X^{i,j}$ represent the feature map at depth i and skip pathway stage j. The feature update rule is defined as:

$$X^{i,j} = f\left(X^{i-1,j-1} \oplus X^{i,j-1} \oplus \cdots \oplus X^{i,j-1}\right), \tag{14}$$

where $f(\cdot)$ denotes a convolutional transformation and $\oplus$ indicates concatenation of feature maps from different paths. By densely connecting intermediate nodes, U-Net++ strengthens feature propagation and improves gradient flow during training.

ResNet/DenseNet/VGG U-Net variants, Swin U-Net, and hierarchical SegNet

ResNet/DenseNet/VGG U-Net variants. Several U-Net variants replace the encoder portion with pre-trained deep convolutional networks such as ResNet, DenseNet [24], or VGG, thereby leveraging their powerful feature extraction capabilities. Residual Networks (ResNet) [25] employ residual skip connections within the encoder to mitigate vanishing gradients, facilitating the training of deeper networks. Formally, for an input x, a residual block computes:

$$y = F(x, W) + x, \tag{15}$$

where $F(x, W)$ represents the residual mapping (convolution, batch normalization, and activation) parameterized by weights W.

Dense Convolutional Networks (DenseNet), shown in Fig. 4, employ a dense connectivity pattern in which each layer receives information from all earlier layers in the network. For the l-th layer, the input is formed by concatenating the feature maps produced by the preceding layers:

$$x_l = H_l([x_0, x_1, \ldots, x_{l-1}]), \tag{16}$$

where $H_l(\cdot)$ represents a sequence of operations including convolution, normalization, and a nonlinear activation function. This structure facilitates efficient feature reuse, strengthens gradient propagation, and minimizes redundant feature learning. In DenseUNet architectures, such dense blocks are embedded within the encoder–decoder framework to improve segmentation performance.

VGG-based U-Net encoders [26] employ stacks of 3×3 convolutional layers with increasing depth. Their simplicity and consistent receptive field make them effective for hierarchical feature extraction in medical imaging tasks.

Swin U-Net. The Swin U-Net architecture [27] integrates the Swin Transformer as the encoder backbone, enabling the model to capture long-range dependencies through hierarchical self-attention mechanisms. Unlike global self-attention, which incurs quadratic complexity, the Swin Transformer divides the image into non-overlapping windows and computes self-attention within each window. To improve cross-window interaction, the windows are shifted at alternating layers.

Formally, given a sequence of patch embeddings $X \in \mathbb{R}^{N \times d}$, the self-attention within a window is computed as:

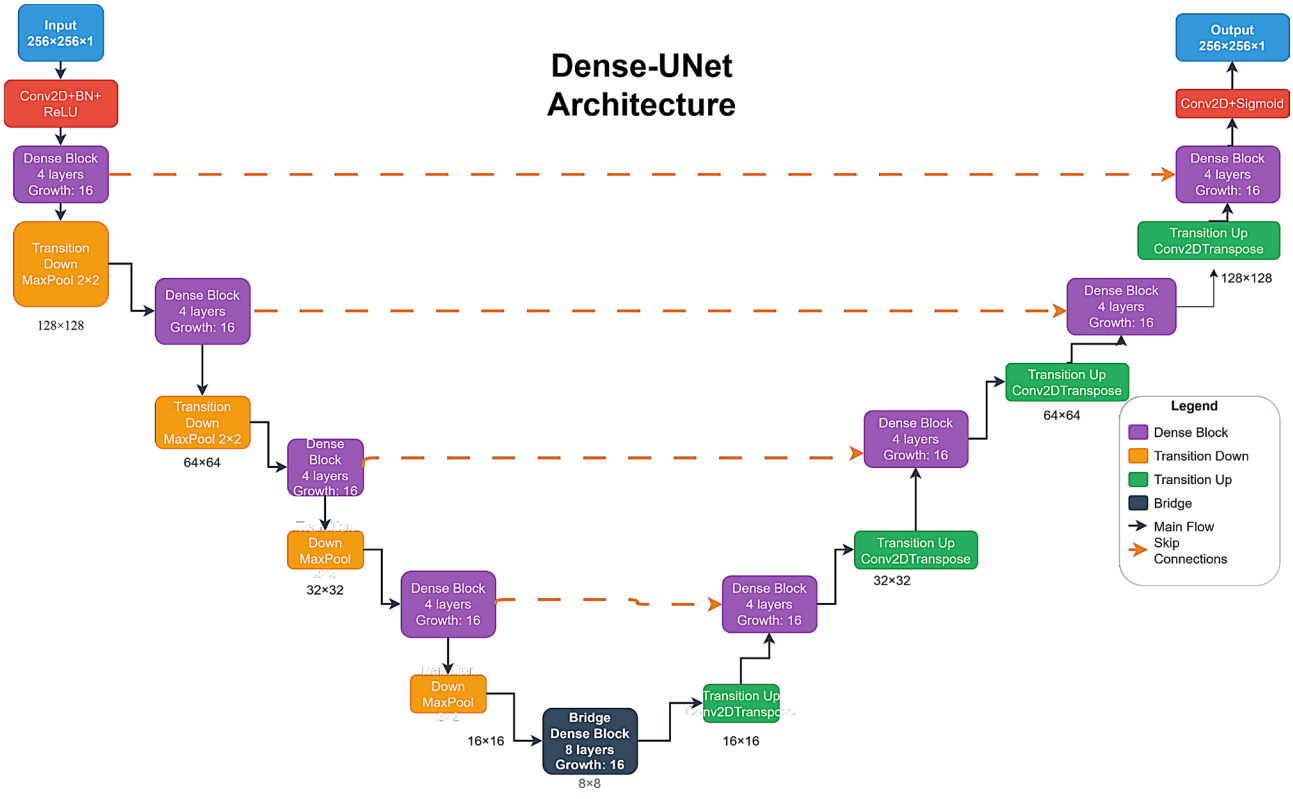


Fig. 4 Architecture of Dense UNet

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V, \quad (17)$$

where Q, K, V are the query, key, and value projections of X, d is the dimension, and B is the learnable relative positional bias. By hierarchically merging patches, the Swin U-Net achieves multi-scale contextual representation, which is beneficial for medical image segmentation.

Hierarchical SegNet. SegNet [28] is an encoder-decoder architecture that employs pooling indices from the encoder during the decoder's upsampling process, ensuring better spatial localization. Hierarchical SegNet extends this framework by introducing multi-scale or hierarchical representations at different depths of the encoder-decoder pipeline. This design improves the ability to segment structures at multiple scales, which is particularly valuable for medical imaging applications such as lung boundary delineation. Although no architectural figures are included for these models, their mathematical formulations and structural descriptions highlight their contributions to robust feature extraction and effective segmentation.

Training and evaluation

The models were trained using the AdamW optimizer with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-5}. A hybrid loss function combining Dice loss and Binary Cross-Entropy (BCE) loss was employed to balance region overlap accuracy and pixel-level classification reliability. Training was performed for 100 epochs with a batch size of 16. To optimize convergence and prevent overfitting, the following callbacks were incorporated: (i) `ModelCheckpoint` to save the best-performing model based on the validation Dice coefficient, (ii) `ReduceLROnPlateau` to reduce the learning rate by a factor of 0.5 if no validation Dice improvement was observed for 5 consecutive epochs, and (iii) `EarlyStopping` to halt training if the validation Dice did not improve for 15 epochs, restoring the best weights. Model performance was evaluated on the validation set using multiple complementary metrics:

- **Dice Coefficient:** A widely used overlap measure that computes the harmonic mean of precision and recall. It ranges from 0 (no overlap) to 1 (perfect overlap).

$$\text{Dice} = \frac{2 \times |P \cap G|}{|P| + |G|}, \quad (18)$$

where P and G denote the predicted and ground truth segmentation masks.

- **Intersection over Union (IoU):** Also known as the Jaccard Index, IoU measures the ratio of intersection to union between P and G.

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|}. \quad (19)$$

- **Hausdorff Distance:** Measures the maximum boundary error by calculating the greatest distance of a point on one boundary set to the nearest point on the other. It reflects worst-case segmentation discrepancy.
- **Average Symmetric Surface Distance (ASSD):** Computes the average symmetric distance between the boundaries of P and G:

$$\text{ASSD}(P, G) = \frac{1}{|B_P| + |B_G|}\left(\sum_{p \in B_P} d(p, B_G) + \sum_{g \in B_G} d(g, B_P)\right), \quad (20)$$

where B_P and B_G are boundary points of prediction and ground truth, and $d(\cdot)$ denotes the Euclidean distance to the closest boundary point.
- **Mean Absolute Error (MAE):** Quantifies the average magnitude of pixel-wise errors:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |P_i - G_i|, \quad (21)$$

where n is the total number of pixels, and P_i, G_i are the predicted and ground truth pixel values.

These metrics collectively provide a comprehensive evaluation framework, assessing both volumetric overlap (Dice, IoU, Accuracy) and boundary precision (Hausdorff, ASSD), as well as pixel-level intensity agreement (MAE).

Experimental results

This section presents the experimental results. The comparative evaluation of ten deep learning architectures for lung segmentation from chest X-rays provides valuable insights into their relative strengths and limitations. Performance was assessed using Dice Coefficient, Intersection over Union (IoU), Mean Absolute Error (MAE), Hausdorff Distance, and Average Symmetric Surface Distance (ASSD), which together capture both overlap accuracy and boundary precision. As shown in Table 1 and Fig. 5, the quantitative performance of CNN-based segmentation models on chest X-ray images demonstrates that the Attention-UNet outperforms the standard UNet

Table 1 Quantitative performance comparison of CNN-based segmentation models on chest X-ray images. Best results are highlighted in bold

Model No.	Model	Dice	IoU	MAE	Hausdorff	ASSD
1	U-Net	0.9797	0.9601	2.6551	15.6619	1.6229
2	Attention U-Net	0.9816	0.9638	2.7617	16.2770	1.5956
3	Double U-Net	0.9843	0.9691	3.0146	16.8545	1.5844
4	U^2-Net	0.9584	0.9202	**1.7015**	13.5744	2.0572
5	VGG-UNet	0.9840	0.9685	3.1247	13.1044	1.5707
6	UNet++	0.9831	0.9668	2.8187	15.1790	1.5941
7	ResNet-UNet	0.9832	0.9669	4.3699	20.9444	2.1083
8	Dense-UNet	**0.9848**	**0.9700**	2.9690	**12.2417**	**1.5279**
9	Swin U-Net	0.9817	0.9641	3.2585	14.7425	1.6715
10	HieraSegNet	0.9799	0.9605	3.3180	29.3995	2.2426

across all evaluation metrics, including Dice Coefficient, IoU, Mean Absolute Error, and Hausdorff Distance.

The baseline U-Net achieved strong results, with a Dice score of 0.9797 and an IoU of 0.9601, confirming its reliability as a foundational model. However, its Hausdorff Distance of 15.66 indicates moderate boundary localization errors, suggesting that while U-Net performs well in general lung delineation, it struggles with finer boundary details.The Attention U-Net improved slightly on overlap metrics (Dice: 0.9816, IoU: 0.9638) but introduced a higher MAE (2.76) and Hausdorff Distance (16.27). This implies that while the attention mechanism enhances focus on salient lung regions, it may overfit certain structures, leading to less consistent pixel-level accuracy.The Double U-Net delivered one of the highest Dice scores (0.9843) and IoU (0.9691), reflecting excellent segmentation overlap. However, its higher MAE (3.01) and Hausdorff Distance (16.85) suggest boundary inconsistencies. This can be attributed to error propagation across the cascaded U-Net stages, which refine predictions but may amplify small inaccuracies.The U2-Net stood out for achieving the lowest MAE (1.70) and a relatively low Hausdorff Distance (13.57), demonstrating precise pixel-level predictions and good boundary adherence. However, its Dice score (0.9584) and IoU (0.9202) were substantially lower than other models. This indicates that U2-Net captures fine details effectively but struggles with overall region completeness, leading to under-segmentation of lung areas.

The VGG-UNet exhibited strong performance, achieving a Dice of 0.9840 and IoU of 0.9685. Its low Hausdorff Distance (13.10) and competitive ASSD (1.57) highlight its ability to balance overlap accuracy and boundary localization. Nonetheless, its higher MAE (3.12) shows that pixel-wise consistency remains a limitation.The UNet++ and ResNet-UNet both achieved Dice and IoU values above 0.966, confirming their robustness. UNet++ improved feature refinement through dense skip connections, but still produced moderate errors (MAE: 2.81, Hausdorff: 15.18). ResNet-UNet, despite strong overlap metrics, recorded the highest MAE (4.36) and poor boundary performance (Hausdorff: 20.94, ASSD: 2.10), highlighting that residual connections may not generalize well for lung segmentation. The Dense-UNet outperformed all models, achieving the best Dice score (0.9848), IoU (0.9700), and the lowest Hausdorff Distance (12.24) and ASSD (1.52). Its dense connections facilitated feature reuse and improved gradient flow, leading to superior accuracy and boundary precision. The slightly higher MAE (2.96) is a minor trade-off compared to its overall consistency. The Swin U-Net, integrating transformer-based self-attention, achieved a Dice of 0.9817 and IoU of 0.9641. While promising in terms of overlap, its higher MAE (3.25) and ASSD (1.67) suggest computational complexity may hinder precise pixel-level predictions. Finally, HieraSegNet underperformed compared to other models, despite a decent Dice score of 0.9799. Its extremely high Hausdorff Distance (29.39) and ASSD (2.24) indicate poor boundary alignment, making it less reliable for clinical applications.

Figures 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 present the training and validation curves of loss, IoU, and Dice coefficient for all ten evaluated models, providing a clear visualization of their learning behavior and convergence trends during training. Among these models, Dense-UNet demonstrates superior performance, with the training loss approaching zero and consistently higher IoU values compared to the other architectures. To further examine the generalization capability of the proposed approach, the best-performing Dense-UNet model was additionally evaluated on a related external dataset [29], where it achieved a Dice score of 0.913, an IoU of 0.902, a mean absolute error (MAE) of 3.819, a Hausdorff distance of 13.871, and an average symmetric surface distance (ASSD) of 1.632, indicating its robustness on similar data distributions. Chest X-ray image segmentation has also been investigated by Sedat Orenc et al. [30], who reported a best Dice score of 0.96; in contrast, the proposed Dense-UNet model attains a higher Dice score of 0.98, demonstrating improved segmentation accuracy.

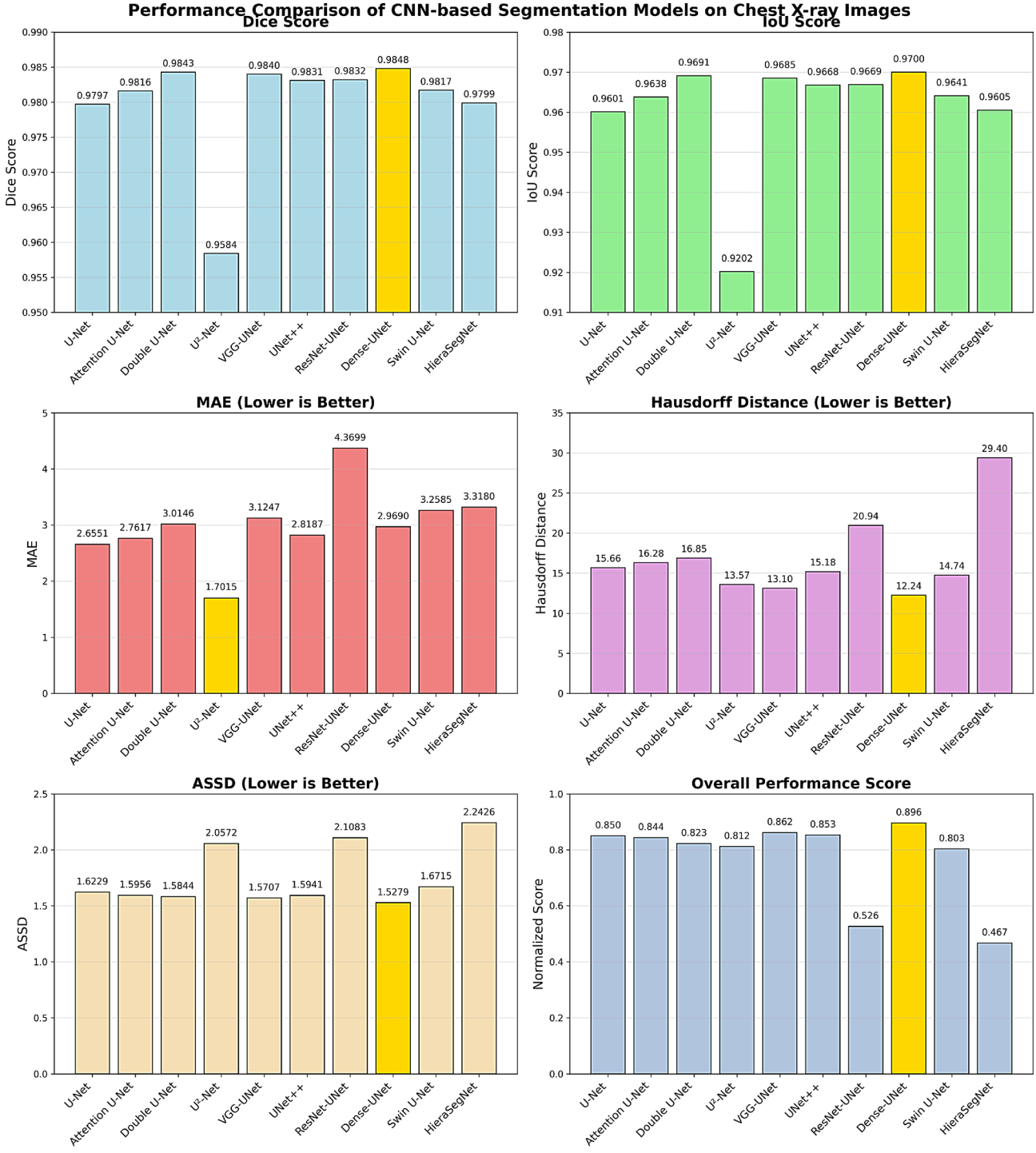


Fig. 5 Performance comparison of CNN models

Discussion

The experimental evaluation highlights important insights into the comparative strengths of different deep learning architectures for lung segmentation in chest radiographs. While all ten models achieved reasonably high Dice and IoU scores, their boundary precision and pixel-level consistency varied significantly, underscoring the trade-offs inherent in architectural design choices. Dense-UNet emerged as the most reliable model, achieving the highest Dice (0.9848) and IoU (0.9700) values, along with the lowest Hausdorff Distance and ASSD. This superior performance can be attributed to its dense connectivity, which promotes feature reuse, mitigates vanishing gradient issues, and enhances boundary delineation.

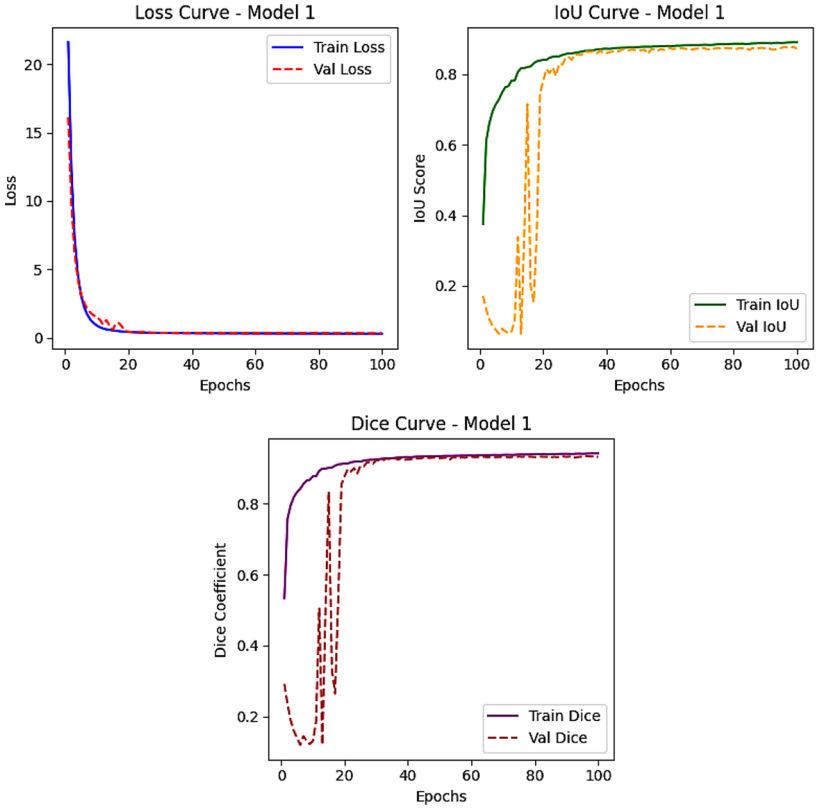


Fig. 6 Training and validation curves for loss, IoU, Dice Coefficient - UNet

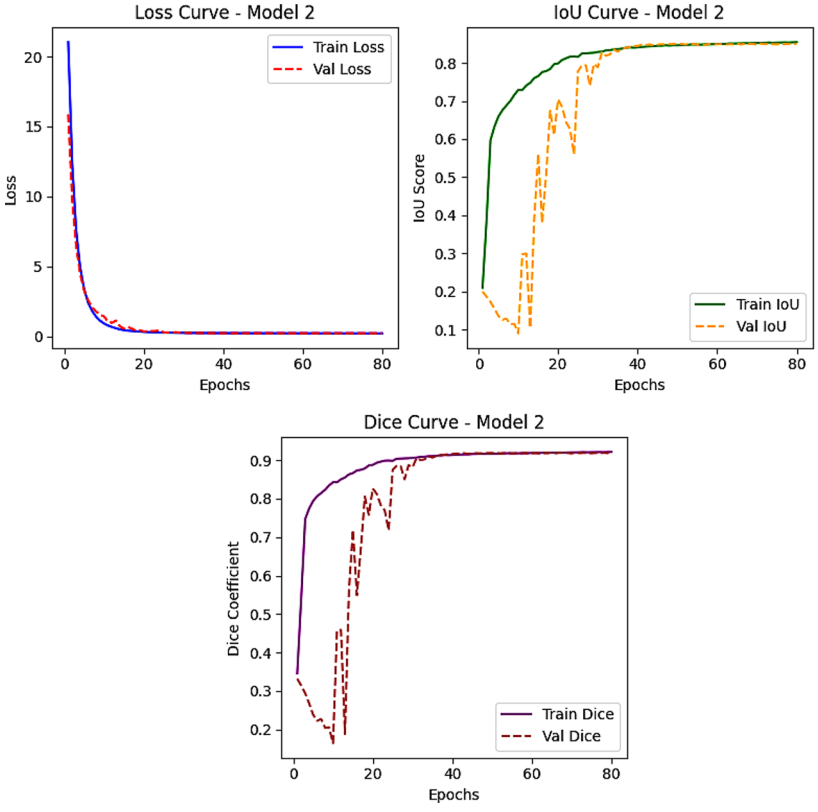


Fig. 7 Training and validation curves for loss, IoU, Dice Coefficient - Attention-UNet

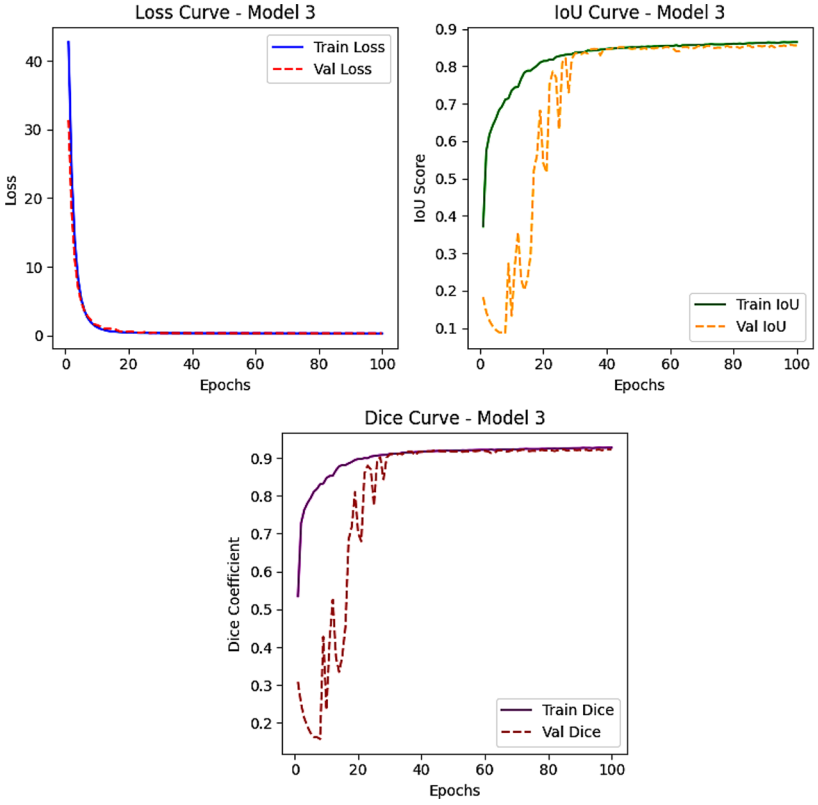


Fig. 8 Training and validation curves for loss, IoU, Dice Coefficient - Double UNet

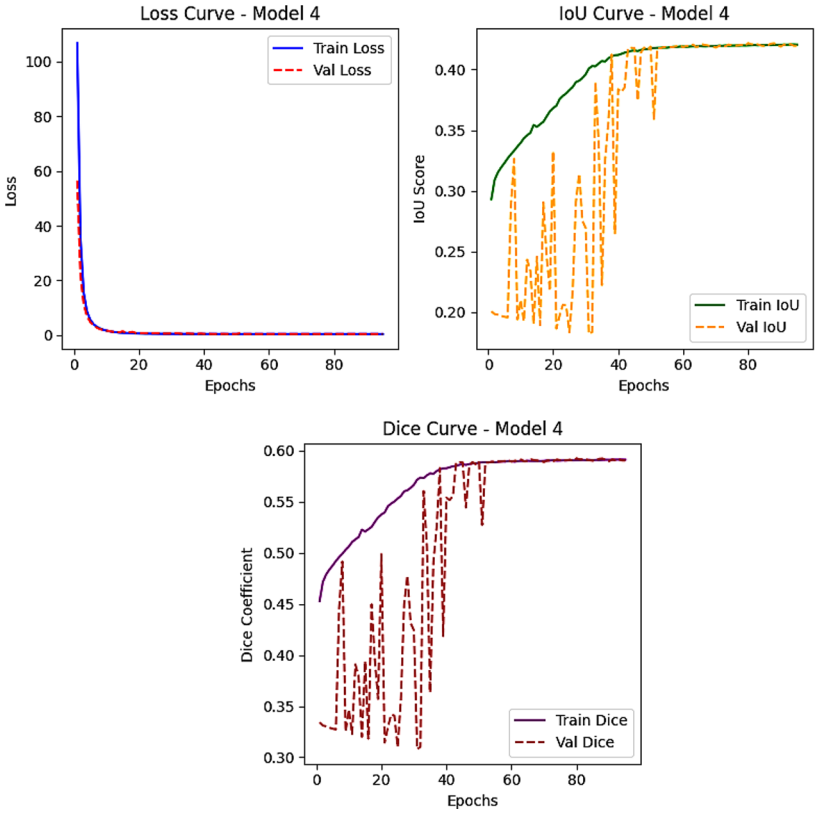


Fig. 9 Training and validation curves for loss, IoU, Dice Coefficient - U2-Net

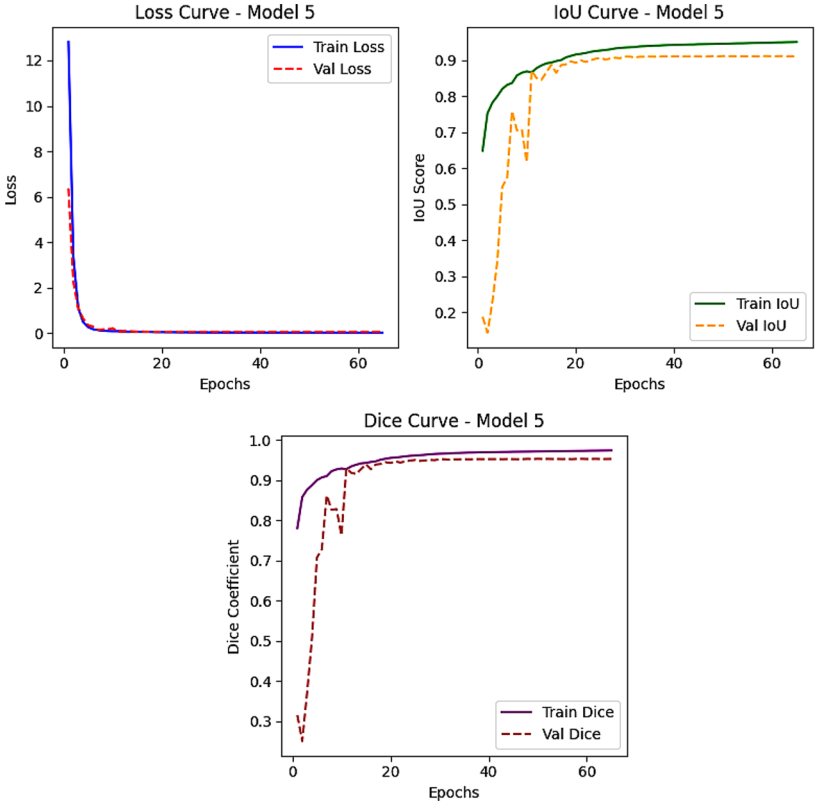


Fig. 10 Training and validation curves for loss, IoU, Dice Coefficient - VGG-UNet

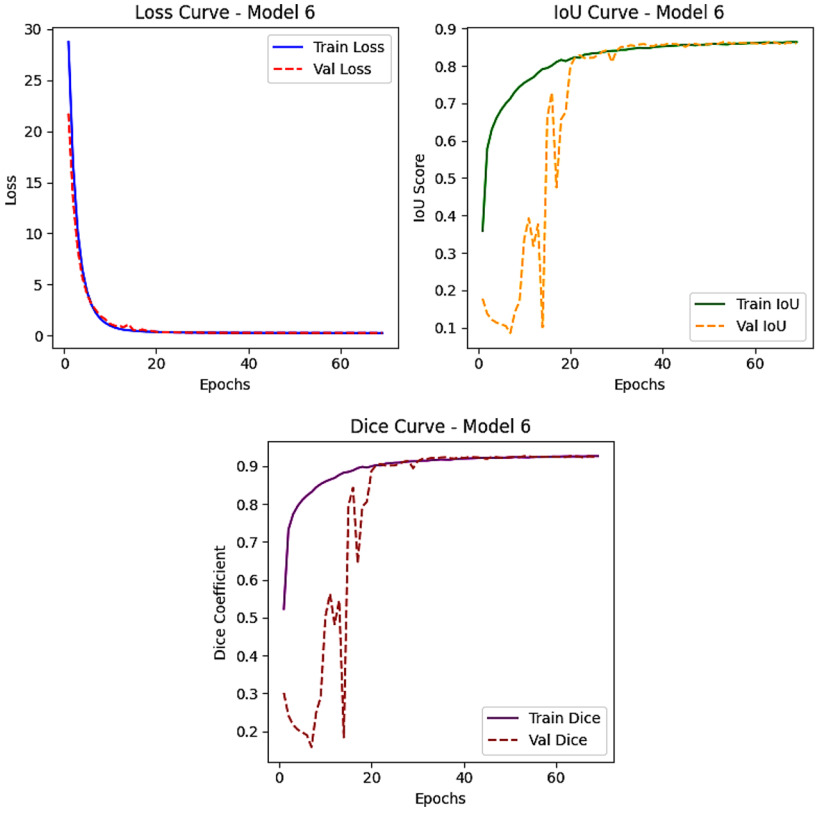


Fig. 11 Training and validation curves for loss, IoU, Dice Coefficient - UNet++

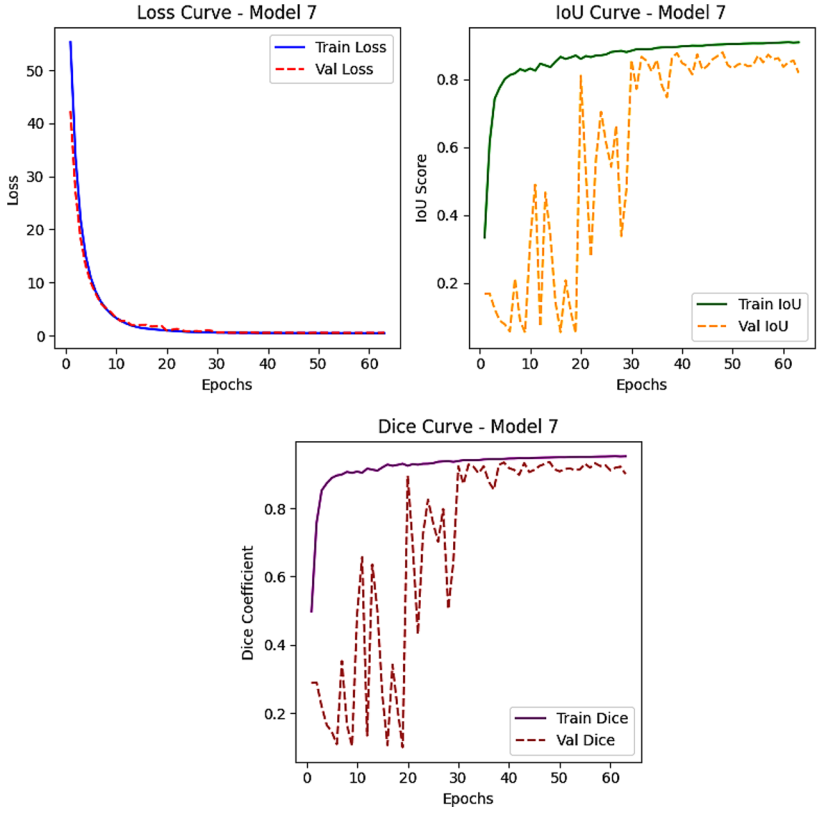


Fig. 12 Training and validation curves for loss, IoU, Dice Coefficient - ResNet-UNet

Its results demonstrate that dense feature propagation is particularly effective for medical image segmentation tasks requiring both global context and local detail.Other architectures showed notable strengths but also key limitations. U2-Net achieved the lowest Mean Absolute Error, indicating highly accurate pixel-level predictions, but its relatively low Dice and IoU scores suggest incomplete coverage of lung regions. Similarly, Double U-Net and VGG-UNet achieved competitive overlap metrics but suffered from higher MAE, reflecting inconsistencies in fine-grained segmentation. ResNet-UNet, despite strong overlap measures, exhibited the poorest boundary localization, highlighting that residual connections may not always generalize well in this domain. HieraSegNet performed least effectively overall, particularly in terms of boundary errors, limiting its clinical utility. Overall, these findings establish Dense-UNet as the most suitable architecture for automated lung segmentation, providing a strong backbone for AI-driven diagnostic pipelines in infectious respiratory diseases such as pneumonia and tuberculosis. If we also compare this with the latest work by Orenc S et al. [30] who have applied several variations of the UNet model and UNet9 achieved the best IoU 0.96, Our best result surpasses this by 0.01 and achieved 0.97.

Conclusion and future scope

In this study, we systematically evaluated multiple deep learning-based architectures for automated lung segmentation in chest radiographs, a critical preprocessing step for the diagnosis of respiratory diseases such as pneumonia and tuberculosis. Our findings demonstrate that Dense-UNet consistently outperformed competing models across a range of evaluation metrics, achieving superior accuracy and boundary precision. By providing reliable isolation of lung regions, Dense-UNet establishes a strong foundation for the development of AI-driven diagnostic pipelines targeting infectious and non-infectious pulmonary conditions. The integration of such segmentation frameworks into clinical workflows can support faster, more consistent, and scalable diagnostic solutions, particularly in resource-limited healthcare settings where the burden of infectious respiratory diseases is high. Future research will focus on validating these approaches across diverse datasets, incorporating multi-class infection-specific segmentation, and advancing toward real-time clinical deployment.

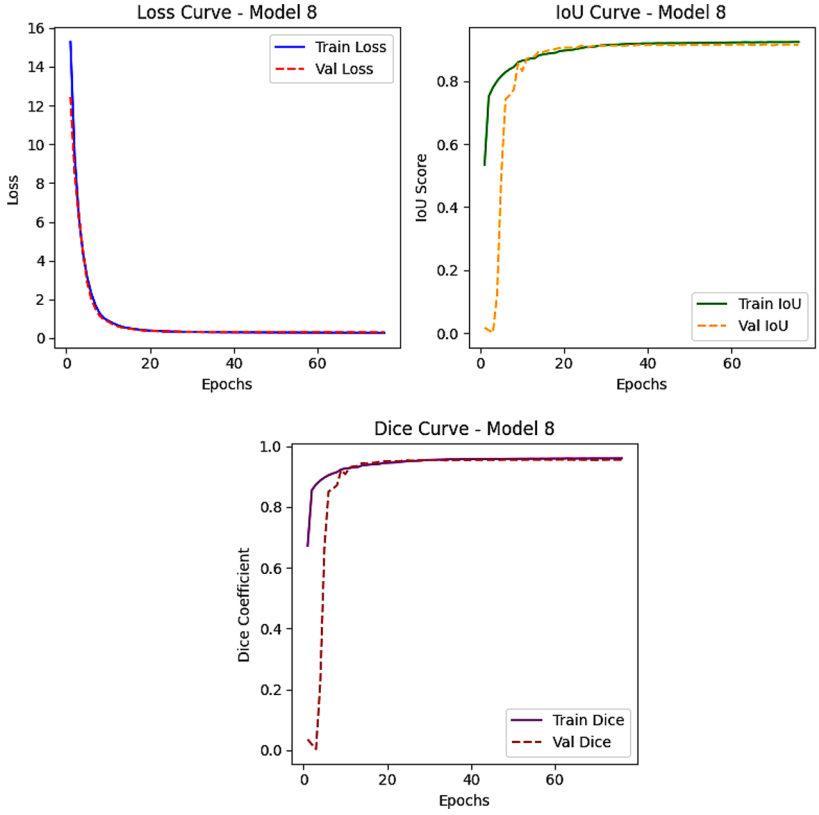


Fig. 13 Training and validation curves for loss, IoU, Dice Coefficient - Dense-UNet

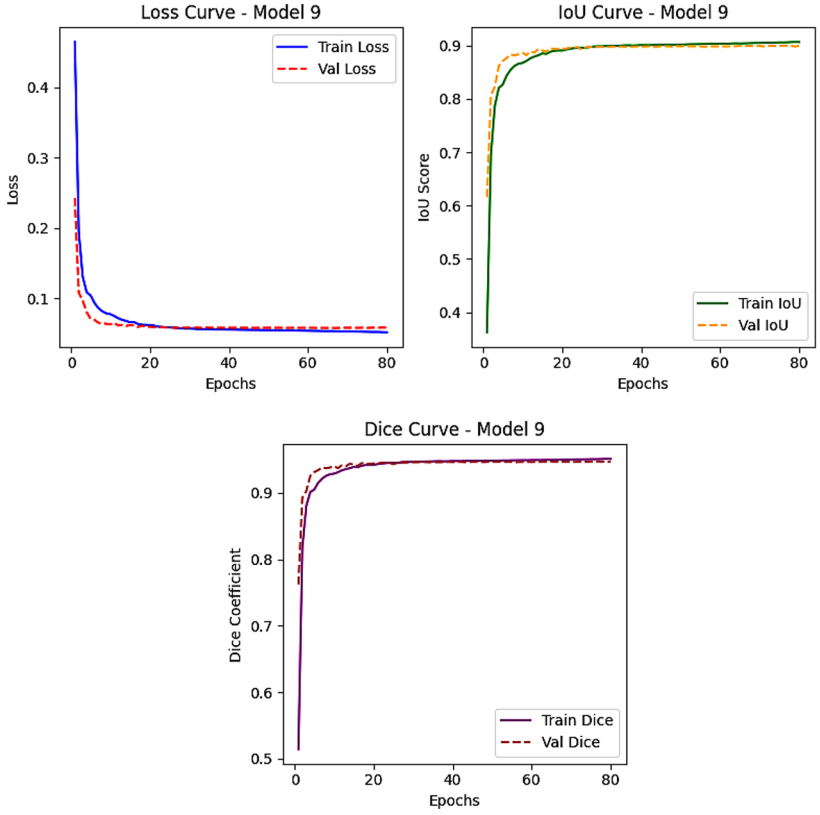


Fig. 14 Training and validation curves for loss, IoU, Dice Coefficient - Swin U-Net

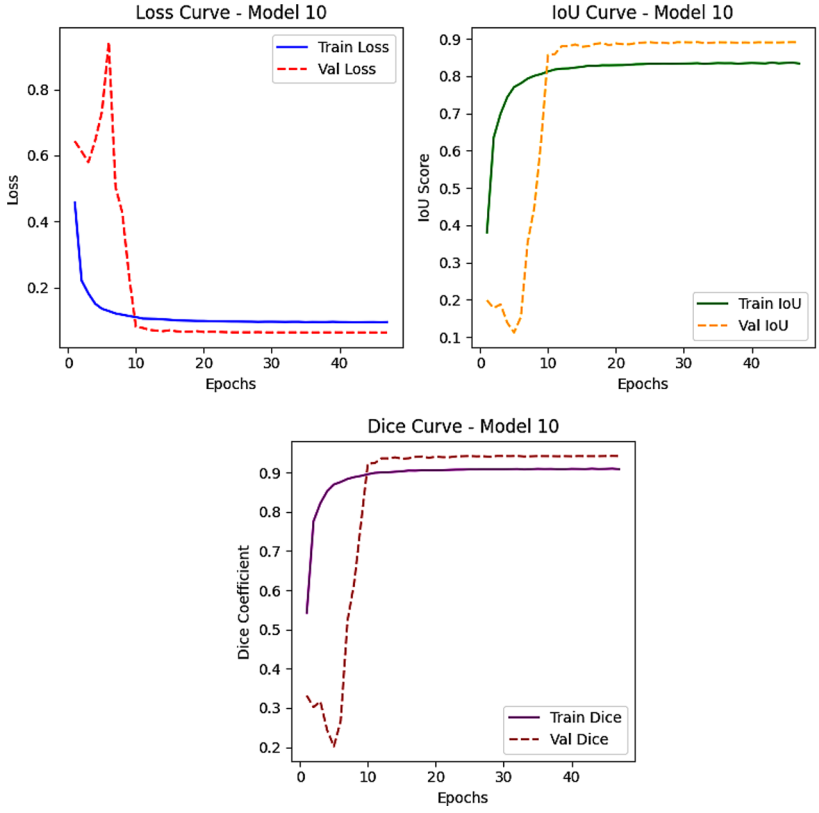


Fig. 15 Training and validation curves for loss, IoU, Dice Coefficient - HieraSegNet

Supplementary Information

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Supplementary Material 1

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Authors contribution
A.K.D, A.J, S.K, M.K, A.P worked on concept, design, performed the experiment, and wrote the manuscript, while A.P, M.A.A, J.K, M.N.A reviewed and edited the manuscript, validated the result.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
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Competing interests
The authors declare no competing interests.

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